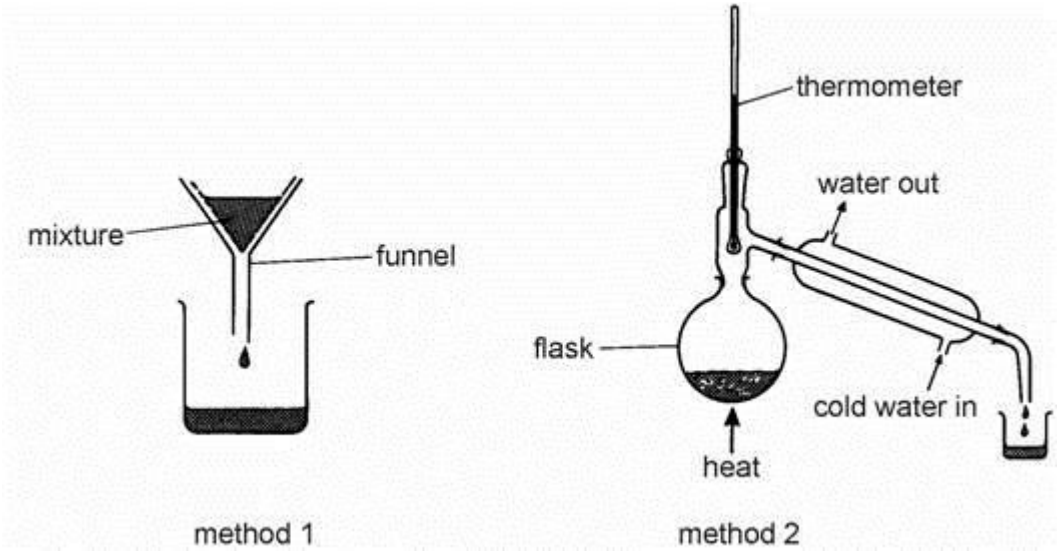
Q1. The following diagrams show two methods of separating substances.



(a) What is the name of each method?

Method 1 is .....

1 mark

Method 2 is .....

1 mark

(b) (i) Tick one box to show which of the mixtures can be separated by method 1.

sugar and salt ☐

sand and water ☐

dissolved salt and water ☐

sand and iron filings ☐

sugar and salt, both dissolved in water ☐

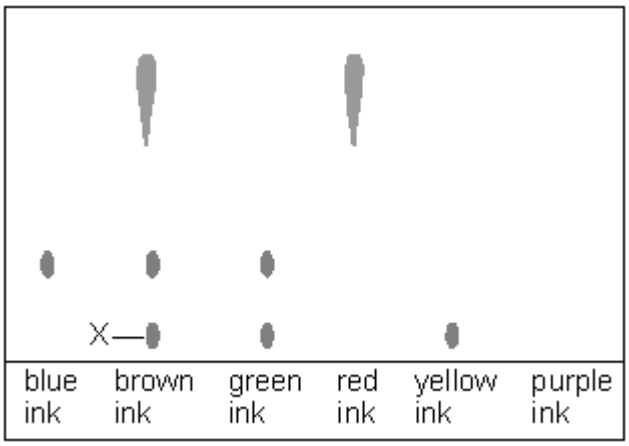
1 mark

(ii) From the list give a mixture which can be separated by method 2 but not by method 1.

.....

1 mark

(c) Chromatography was used to analyse some soluble inks. The results are shown below.



(i) A purple ink is a dissolved mixture of the red dye and the blue dye. On the right of the diagram draw the pattern you would expect to see for purple ink.

1 mark

(ii) Which **three** inks contain only one dye?

.....

1 mark

(iii) What colour is spot **X**?

.....

1 mark

Maximum 7 marks

Mark schemes

Q1.

(a) filtration **or** filtering

1

distillation **or** distilling

1

(b) (i) sand and water

if more than one box is ticked award no mark

1

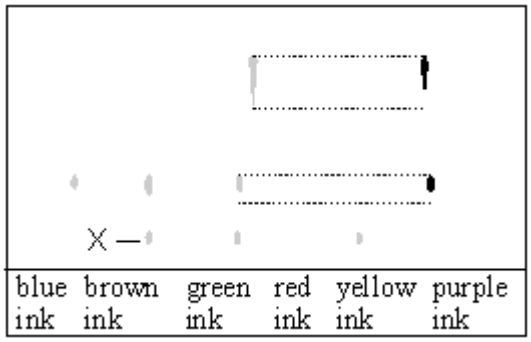
(ii) dissolved salt and water

1

(c) (i) the correct positions of both red and blue spots are required for the mark

if more than two spots are drawn award no mark

*the sizes of the spots need **not** be the same as those in the question*



1

(ii) blue, yellow, red

*all **three** must be correct for the one mark*

1

(iii) yellow

1

[7]